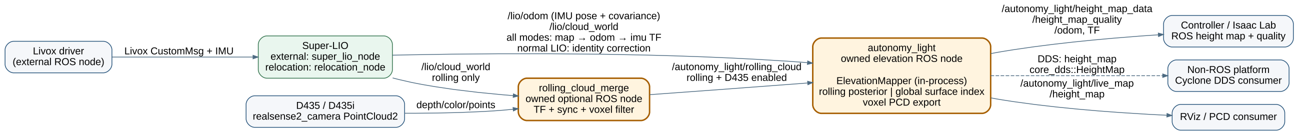
autonomy_light software architecture

Super-LIO integration and elevation mapping · 2026-08-03

> **Design decision.** SLAM and relocalization are Super-LIO responsibilities. This package owns the elevation node and an optional, isolated D435 rolling-cloud merge node. Super-LIO remains the only SLAM backend.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry with ESKF pose covariance, Scan Context + GICP + iSAM2 full SLAM, saved-map relocation, and continuous `map → odom → imu` TF.
`rolling_cloud_merge`	this package, optional	Synchronizes one D435/D435i PointCloud2 to a LiDAR cloud, transforms it into `odom|map`, and voxel filters it.
`autonomy_light`	this package	Base pose/TF, rolling or global elevation map, quality output, live PCD and PCD save.

Data and TF contract

Direction	Topic	Meaning
input	`/lio/odom`	Super-LIO IMU pose and 6×6 ESKF covariance; exactly converted to `base_link`.
input	`/lio/cloud_world`	Registered cloud in the current global frame.
optional input	`/camera/.../depth/color/points`	One D435/D435i cloud; used only by the merge node.
intermediate	`/autonomy_light/rolling_cloud`	Time-aligned, global-frame LiDAR+D435 cloud for rolling elevation.

output	`/autonomy_light/height_map_data`	Controller distance grid.
output	`/autonomy_light/height_map_quality`	Per-cell validity, variance, age.
output	`/autonomy_light/live_map`	Sparse voxel PCD for RViz and map saving.
optional output	`height_map`	Direct Cyclone DDS `core_dds::HeightMap` payload for a non-ROS peer.

TF is `map → odom → imu → base_link → base_link_gravity`, with static `imu → base_link → lidar_link`. Super-LIO publishes identity `map → odom` before its first LiDAR update, then owns the correction; this runtime owns the two physical static links. Normal LIO keeps that correction at identity. A saved map runs Super-LIO `relocation_node`. After its NDT→ICP registration succeeds, Super-LIO continuously publishes `map → odom → imu`; the correction is only recalculated at the configured low global-registration rate. Mapping runs the pose graph in Super-LIO. There is no `world` frame or duplicate SLAM backend in this package.

`/lio/odom` is the IMU state. The runtime composes `T_base_imu = T_base_lidar · inverse(T_imu_lidar)` from `target_to_lidar_*` and `imu_from_lidar_*`, then applies `T_global_base = T_global_imu · inverse(T_base_imu)`. Its managed Super-LIO process receives the same `T_imu_lidar` calibration.

Elevation source selection

Configuration	Algorithm	Use
`height_map.source: rolling`	Bounded Bayesian ground/upper posterior with measurement variance, aging and outlier gate.	Default for control and sim-to-real.
`height_map.source: global`	Persistent, PCD-derived sparse surface index.	Repeatable terrain reference from a saved or accumulated map.

Both sources use the same output message. `valid=0` means the distance value is an unknown placeholder, not measured flat ground; Isaac Lab should consume the quality channels.

With `rolling_merge.enabled: true`, the launcher starts the optional merge node. It accepts only a D435 sample within the configured synchronization window, uses TF at that camera timestamp, and passes a LiDAR-only cloud through if TF or a matching camera sample is unavailable.

The floor origin uses a local `z=ax+by+c` ground-plane fit with Huber residual weights; `c` is the floor directly below `base_link`. It rejects fits with too few cells or a slope above the configured limit, then falls back to the nearby 20th-percentile ground height. This is robust to isolated low depth returns while retaining slope tracking.

Each rolling cloud is matched to its timestamped odometry covariance. Sensor range and mounting calibration are added to the Jacobian-projected pose covariance; XY uncertainty splats the point across nearby cells. Over-limit uncertainty rejects the cloud. The merge node keeps a `sensor_id` field so LiDAR and D435 retain independent measurement models.

Transport selection

`height_map_output.transport` selects `ros2`, `cyclone_dds`, or `both`. The direct DDS option uses the installed IDL `core_dds::HeightMap { sequence<float> data; }`, configurable domain and topic, and best-effort keep-last delivery. Its flat, row-major data sequence has exactly the same values and unknown convention as ROS `HeightMap.data`; geometry and quality remain ROS-message metadata. `both` is the migration/validation mode.

Removed ownership

- Duplicate local pose graph, Scan Context loop closure, and map reconstruction
- FPFH/GICP initial localization and runtime map correction outside Super-LIO
- Secondary LiDAR merger and custom ROS-domain bridge executors
- Separate raw/refined/ROI map pipelines and duplicated path subscription

These removals make the state path linear: Super-LIO localizes; one node maps elevation.